

Experiment

Aim

To create a queue and perform Enqueue, Dequeue, Peek and Traverse operations using an array.

Algorithm

1. Start.
2. Declare an array for the queue.
3. Set `front = 0` and `rear = -1`.
4. For Enqueue, increase `rear` and insert the element.
5. For Dequeue, remove the element from `front` and increase `front`.
6. For Peek, display the front element.
7. For Traverse, display all elements from front to rear.
8. Stop

Source Code

```
#include <iostream>

using namespace std;

int main()
{
    int queue[5];

    int front = 0, rear = -1;

    int choice, value;

    do
    {
        cout << "\n1. Enqueue";
```

```
cout << "\n2. Dequeue";

cout << "\n3. Peek";

cout << "\n4. Traverse";

cout << "\n5. Exit";


cout << "\nEnter choice: ";

cin >> choice;


switch(choice)
{
    case 1:
        if(rear == 4)
        {
            cout << "Queue is full";
        }
        else
        {
            cout << "Enter element: ";

            cin >> value;


            rear++;

            queue[rear] = value;
        }

        break;
```

case 2:

```
if(front > rear)
{
    cout << "Queue is empty";
}
else
{
    cout << "Deleted element: " << queue[front];
    front++;
}
break;
```

case 3:

```
if(front > rear)
{
    cout << "Queue is empty";
}
else
{
    cout << "Front element: " << queue[front];
}
break;
```

case 4:

```
if(front > rear)
{
    cout << "Queue is empty";
}
else
{
    cout << "Queue: ";

    for(int i = front; i <= rear; i++)
    {
        cout << queue[i] << " ";
    }
}
break;
```

case 5:

```
cout << "Exit";

break;
```

default:

```
cout << "Invalid choice";
}
```

```
    } while(choice != 5);

    return 0;

}
```

Output

```
• PS C:\Users\ABHISHEK\Downloads\c++work> g++ dsqueue.cpp -o queue.exe
• PS C:\Users\ABHISHEK\Downloads\c++work> ./queue.exe

1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 1
Enter element: 6

1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 1
Enter element: 7

1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 1
Enter element: 8

1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 3
Front element: 6
```

```
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 4
Queue: 6 7 8
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 2
Deleted element: 6
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 4
Queue: 7 8
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 5
Exit
PS C:\Users\ABHISHEK\Downloads\c++work>
```

PS C:\Users\ABHISHEK\Downloads\c++work> ./queue.exe

1. Enqueue

2. Dequeue

3. Peek

4. Traverse

5. Exit

Enter choice: 2

Queue is empty

1. Enqueue

2. Dequeue

3. Peek

4. Traverse

5. Exit

Enter choice: 1

Enter element: 3

1. Enqueue

2. Dequeue

3. Peek

4. Traverse

5. Exit

Enter choice: 1

Enter element: 4

1. Enqueue

2. Dequeue

3. Peek

4. Traverse

5. Exit

Enter choice: 1

Enter element: 6

```
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 1
Enter element: 2
```

```
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 1
Enter element: 7
```

```
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: 1
Queue is full
1. Enqueue
2. Dequeue
3. Peek
4. Traverse
5. Exit
Enter choice: █
```